Network Hardware Devices & Infrastructure Assignment

Comprehensive Long-Form Technical Report & Solution Guide (Tasks 1 to 4)

Task 1: Core Functions & Hardware Ports of Network Devices

In modern home and enterprise Wi-Fi environments, multiple specialized networking devices work in tandem to connect wireless clients, switch local Ethernet traffic, route IP packets, and secure network boundaries. Below is a detailed technical matrix comparing their primary functions, operational layers, typical hardware ports, and real-world roles:

Device Name	OSI Layer	Main Primary Function	Typical Physical Hardware Ports
Network Switch	Data Link (Layer 2)	Connects multiple wired LAN devices using MAC address tables (`CAM table`) to forward data frames directly to destination ports without collision.	4/8/16/24/48 x RJ45 Ethernet LAN Ports, SFP/SFP+ Optical Uplink Ports, Management Console Port (RJ45/DB9), DC Power Jack.
Router	Network (Layer 3)	Connects different networks (e.g., LAN to WAN), routes IP packets based on logical IP addresses, performs NAT (Network Address Translation), and runs DHCP services.	1 x RJ45 WAN (Internet) Port, 4 x RJ45 LAN Ports, USB Port (for storage/ modems), Reset Button, DC Power Port.
Hub	Physical (Layer 1)	Legacy multiport repeater that broadcasts incoming bits to ALL connected ports, creating a single shared collision domain (half-duplex).	4 to 8 x RJ45 Ethernet Ports, Power Connector (No console or management ports available).
Firewall	Layers 3 to 7 (NGFW)	Inspects incoming/outgoing traffic, enforces security rules, filters malicious packets, blocks unauthorized access, and maintains stateful inspection tables.	Dedicated WAN/LAN Ports, DMZ (Demilitarized Zone) Port, Serial/USB Console Port for CLI management, Redundant Power Supplies.
Access Point (AP)	Data Link / Physical	Converts wired Ethernet signals into wireless Radio Frequency (RF) signals (Wi-Fi 5/6/7) to connect wireless smartphones and laptops to a wired LAN.	1 x Gigabit RJ45 PoE (Power over Ethernet) LAN Port, External RP-SMA Antenna Connectors, Reset Button, DC Power Input.

Task 2: Hardware Device Labeling & Physical Port Identification

Physical networking hardware features specialized interfaces designed for connectivity, power, out-of-band management, and signal radiation. Below is the structural port-level identification for three core networking devices:

Device 1: Wireless Wi-Fi Router (Home Gateway)

- **Ethernet WAN Port (Blue/Yellow RJ45):** Connects to the ISP optical fiber modem (ONT) to receive public internet IP via PPPoE or DHCP.
- **Ethernet LAN Ports (4 x RJ45):** Connects wired local devices like PCs, Smart TVs, or gaming consoles.
- **Antenna Connectors (Omnidirectional RP-SMA/Fixed):** 2.4 GHz and 5 GHz dipoles for radiating Wi-Fi signals.
- **Power Interface:** 12V DC power barrel jack with ON/OFF rocker switch.

Device 2: Enterprise Managed Ethernet Switch

- **Ethernet Ports (24 x Gigabit RJ45):** Connects workstation PCs, IP phones, and wireless APs with PoE support.
- **Console Port (RJ45 / Serial RS-232):** Connects directly to an admin laptop via rollover cable for initial CLI command setup (Cisco IOS / Aruba CLI).
- **Power Interface:** Standard C14 AC power inlet (100V-240V internal PSU).

Device 3: Enterprise Wireless Access Point (Ceiling Mount AP)

- **PoE Ethernet Port (1 x RJ45 PoE+):** Receives both network data and DC power over a single Cat6 cable from a PoE switch.
- **Internal/External Antennas:** High-gain directional patch or omni antennas housed under a sleek dome.
- **Console Port (Micro-USB / Serial):** Low-profile port for local hardware diagnostics and recovery.

Task 3: Real-World App Role Analysis: Ordering Food on Zomato

When you open the **Zomato app** on your smartphone, select a restaurant, and place a food order, data flows seamlessly through multiple home and cloud networking devices:

> 📱 **Zomato Food Ordering Network Flow:**
>
> 1. **Wireless Access Point (AP):** Your smartphone converts your Zomato order request into 5 GHz Wi-Fi radio signals. The local Access Point catches these RF signals, converts them into Ethernet data frames, and forwards them to the home network.
>
> 2. **Router:** The home router receives the data frame, assigns your private IP address to a public IP using **NAT (Network Address Translation)**, determines the optimal path across the internet using Layer 3 IP routing, and securely pushes the order to Zomato's cloud backend servers!

Task 4: Troubleshooting College Wi-Fi Disconnections & Device Upgrade Solution

In a college environment with hundreds of students simultaneously trying to access online lectures and assignments, slow speeds and frequent disconnections are common infrastructure symptoms.

Root Cause Analysis (Device Capacity & Channel Overcrowding):

The primary issue is likely **Access Point Overload & Co-Channel Interference**. Consumer-grade or outdated Access Points (Wi-Fi 4/5) can only handle 20–30 simultaneous client connections per radio before their CPU,

memory, and channel bandwidth become saturated. When 100+ students enter a hall, the AP's airtime capacity collapses, resulting in high packet loss and frequent client disconnections.

Recommended Upgrade Solution:

> 💡 **Upgrading to Enterprise Wi-Fi 6 / 6E Access Points (with MU-MIMO & OFDMA):**
> Upgrading to modern **Enterprise Wi-Fi 6 (802.11ax) Access Points** (such as Cisco Catalyst or Aruba APs) solves this issue because Wi-Fi 6 introduces **OFDMA (Orthogonal Frequency-Division Multiple Access)** and **8x8 MU-MIMO**. This allows a single AP to communicate with up to 200+ active devices simultaneously without channel congestion, eliminating disconnections and delivering stable, high-speed connectivity across the campus!